

# **DIPLOMA IN IT IN NETWORK DESIGN AND ADMINISTRATION**

**FRIDAY PRACTICAL**

**NETWORKS 731**



**CTU**  
training solutions

Private  
Tertiary  
Education

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## ASSESSMENT RULES AND REGULATIONS

### Instructions to Students:

- This project consists of 10 weekly tasks, each with specific deliverables. Students must complete all tasks as outlined in the rubric.
- Students must submit all weekly evidence to GitHub as they progress through the project.
- By 12 May 2026, students must download their entire GitHub repository as a ZIP folder and upload it on Campus Online for assessment.
- Type all documents and reports in Microsoft Word and convert them to PDF before uploading.
- Use Calibri font, size 11, for all written submissions.
- Complete the Declaration of Authenticity and include it in the submission.
- Use the Harvard referencing method and provide a minimum of five (5) references. Three (3) marks will be deducted from the final total if the work is not properly referenced.
- Include a Table of Contents in your submission. Failure to do so will result in a deduction of two (2) marks from the final total Mark.

**You are required to make use of the Harvard Referencing method. Not referencing your work constitutes plagiarism.**

## DEFERMENT APPLICATION

Your deferment application will only be considered if submitted:

Within a **5-day period** after the initial assessment was conducted, and once the following required documentation is uploaded via CampusOnline:

1. Affidavit signed by a Commissioner of Oaths or the South African Police, **clearly stating the reason.**
2. Doctor's note from a registered health professional.
3. Death Certificate of an immediate family member.

To submit your deferment application:

1. Log into CampusOnline.
2. Click on "Current Courses" and navigate to "Click here to apply for an assessment deferment."

Before applying, please ensure that all required documents are ready for upload to complete the process successfully.

## DECLARATION OF AUTHENTICITY AND RULES AND REGULATIONS - THEORY

I Ruben Kok  
(FULL NAME)

hereby declare that the contents of this assessment

NET731

is entirely my own work, completed by me without any paraphrasing/copying, or presented as my own work accessed from any **AI Apps, example ChatGPT, Co-pilot, Perplexity, or any other App.**

Signature: 

Date: 2026/03/27

## INTRODUCTION

This project is designed to provide students with practical, hands-on experience in planning, deploying, and managing cloud resources using **Microsoft Azure**. Over a ten-week period, students will engage in a series of tasks that simulate real-world IT project scenarios, including **project planning, network design, virtual machine deployment, identity management, security configuration, monitoring, cost management, and final documentation**.

The primary aim of this assignment is to develop the technical skills and professional practices required to design, implement, and manage cloud-based solutions in a secure and efficient manner. Students will also enhance their **documentation, collaboration, and version control skills** by submitting all evidence and project artifacts to **GitHub** throughout the project.

By the end of this assignment, students will have a fully deployed and documented Azure environment, demonstrating their ability to apply theoretical knowledge to practical cloud computing scenarios while adhering to **project management principles** and best practices.

## LEARNING OUTCOMES:

- Understand how to manage Azure identities and governance.
- Demonstrate the ability to configure and manage virtual networking.
- Demonstrate the ability to configure and manage network traffic
- Demonstrate the ability to implement and manage storage.
- Demonstrate the ability to deploy and manage Azure compute resources.
- Understand how to monitor and maintain Azure resources.

## QUALIFICATION AND ASSESSMENT DETAILS

|                             |                                                    |
|-----------------------------|----------------------------------------------------|
| <b>Programme Title</b>      | DIPLOMA IN IT IN NETWORK DESIGN AND ADMINISTRATION |
| <b>Module Title</b>         | NETWORKS                                           |
| <b>Module Code</b>          | NET731                                             |
| <b>NQF Level</b>            | 7                                                  |
| <b>Formative Assessment</b> | FRIDAY PRACTICAL ASSESSMENT (FA3)                  |
| <b>Hand out date</b>        | 6 March 2026                                       |
| <b>Submission date</b>      | 12 May 2026                                        |
| <b>Total Marks</b>          | 100                                                |
| <b>Examiner</b>             | Reason Katsangu                                    |
| <b>Internal Moderator</b>   | Linda Dube                                         |
| <b>Internal Moderator</b>   | Faith Muwishi and Palesa Nkosi                     |

|                                 |           |
|---------------------------------|-----------|
| <b>Student Name and Surname</b> | Ruben Kok |
| <b>Student Number</b>           | 20240173  |

## AZ-104 Capstone Practical Assignment

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**Purpose:** Design, Deploy, and Secure a Scalable Azure Infrastructure

**Institution:** CTU Training Solutions (Scenario)

**Programme Level:** NQF Level 7

**Duration:** 10 Weeks (6 March – 12 May 2026)

**Submission Method:** GitHub Repository & Campus Online ZIP Upload

**Azure Access:** Azure for Students (\$100 Credit)

### 1 Project Overview

This capstone project simulates the role of an Azure Administrator responsible for designing, deploying, securing, and monitoring a cloud infrastructure using Microsoft Azure. Students must implement Azure services while ensuring the total cost remains within the \$100 Azure for Students credit.

### 2 Learning Outcomes

By completing this project students will demonstrate:

- Ability to design cloud infrastructure
- Implementation of Azure networking
- Identity and access management
- Security configuration using NSGs
- Infrastructure monitoring and cost control
- Professional documentation and version control using GitHub

### 3 Required Tools

- Microsoft Azure Portal
- Azure for Students Subscription
- GitHub Account
- Web Browser (Microsoft Edge or Chrome)
- Diagram tool (Draw.io, Visio, or PowerPoint)



#### 4 GitHub Repository Requirement

Students must create a GitHub repository where all weekly evidence will be submitted. The repository should be named using the following format:

AZ104-Capstone-StudentName

Example: AZ104-Capstone-LindaDube

|                                   |
|-----------------------------------|
| Recommended Repository Structure: |
| Week1_Planning_AzureSetup         |
| Week2_NetworkDesign               |
| Week3_VMDeployment                |
| Week4_IdentityRBACNSG             |
| Week5_SecurityReview              |
| Week6_Monitoring                  |
| Week7_CostManagement              |
| Week8_FinalDocumentation          |
| Week9_FinalPresentation           |

## 5 Weekly Practical Activities & Mark Allocation (100 Marks Total)

### Week 1 (6–12 March) – Project Planning & Azure Setup (10 Marks)

- Define business requirements for CTU Training Solutions
- Identify Azure services required
- Create Azure Resource Group in Azure for Students
- Document region, tags, and configuration
- Upload planning document & setup evidence to GitHub

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### Week 2 (13–19 March) – Network Design (10 Marks)

- Create Virtual Network
- Configure address space and subnets
- Attach Network Security Group to subnets
- Upload network design documentation to GitHub

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### Week 3 (20–26 March) – Virtual Machine Deployment (10 Marks)

- Deploy a B1s Virtual Machine
- Configure networking, public IP, and subnet
- Enable auto-shutdown
- Document VM deployment and upload screenshots to GitHub

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### Week 4 (27 March – 2 April) – Identity Management (8 Marks)

- Create users and groups in Microsoft Entra ID
- Assign initial roles (basic access)
- Document users and group creation in GitHub

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### Week 5 (3–9 April) – Role-Based Access Control & Security Configuration (10 Marks)

- Assign RBAC roles to users in the resource group
  - Configure Network Security Group rules (inbound/outbound)
  - Create an Azure Storage Account
  - Configure containers and upload sample files
  - Set access permissions for users (Blob/File access)
  - Upload RBAC, NSG, and Storage evidence (screenshots and documentation) to GitHub
-

**Week 6 (10–16 April) – Security Review (10 Marks)**

- Open Microsoft Defender for Cloud (Free Tier)
- Review of security recommendations
- Apply fixes (close open ports, enable OS updates)
- Document improvements in GitHub

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**Week 7 (17–23 April) – Monitoring Setup (10 Marks)**

- Enable Azure Monitor on VM and network resources
- Review metrics (CPU, network, disk)
- Document monitoring setup with screenshots

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**Week 8 (24–30 April) – Cost Management (8 Marks)**

- Create Azure budget alerts to track spending
- Monitor usage and adjust resource settings to stay within \$100
- Document cost management process in GitHub

**Week 9 (1–7 May) – Final Documentation (14 Marks)**

- Create architecture diagram
- Document full Azure environment (resources, configurations, security, monitoring)
- Upload diagrams and documentation to GitHub

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**Week 10 (8–12 May) – Final Presentation (10 Marks)**

- Prepare final presentation slides
- Upload slides to GitHub
- Conduct peer review (if required)
- Submit final repository ZIP folder on Campus Online by 12 May 2026

## 6 Submission Instructions

Students must submit all weekly evidence to GitHub as they progress through the project. By 12 May 2026, students must download their entire GitHub repository as a ZIP folder and upload it on Campus Online for assessment.

## 7 Cost Control Guidelines

- Only deploy one B1s Virtual Machine
- Enable auto shutdown on VMs
- Delete unused resources
- Use standard storage tiers
- Avoid paid Azure services such as Azure Firewall
- Estimated total project cost: \$40 – \$50 within the \$100 student credit.

## 8 MARKING RUBRIC

### THE FOLLOWING RUBRIC WILL BE USED TO ASSESS THIS PRACTICAL

| Week | Date               | Tasks / Deliverables                                                                                                                                                                                                                                             | Submission / Evidence                                   | Marks |
|------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-------|
| 1    | 6–12 March         | <ul style="list-style-type: none"> <li>Define business requirements for CTU Training Solutions</li> <li>Identify Azure services required</li> <li>Create Azure Resource Group in Azure for Students</li> <li>Document region, tags, and configuration</li> </ul> | Upload planning document & setup evidence to GitHub     | 10    |
| 2    | 13–19 March        | <ul style="list-style-type: none"> <li>Create Virtual Network</li> <li>Configure address space and subnets</li> <li>Attach Network Security Group to subnets</li> </ul>                                                                                          | Upload network design documentation to GitHub           | 10    |
| 3    | 20–26 March        | <ul style="list-style-type: none"> <li>Deploying a B1s Virtual Machine</li> <li>Configure networking, public IP, and subnet</li> <li>Enable auto-shutdown</li> </ul>                                                                                             | Document VM deployment and upload screenshots to GitHub | 10    |
| 4    | 27 March – 2 April | <ul style="list-style-type: none"> <li>Create users and groups in Microsoft Entra ID</li> <li>Assign initial roles (basic access)</li> </ul>                                                                                                                     | Document users and group creation in GitHub             | 8     |
| 5    | 3–9 April          | <ul style="list-style-type: none"> <li>Assign RBAC roles to users in resource group</li> <li>Configure Network Security Group rules (inbound/outbound)</li> </ul>                                                                                                | Upload RBAC and NSG evidence to GitHub                  | 10    |
| 6    | 10–16 April        | <ul style="list-style-type: none"> <li>Open Microsoft Defender for Cloud (Free Tier)</li> <li>Review security recommendations</li> <li>Apply fixes (close open ports, enable OS updates)</li> </ul>                                                              | Document improvements in GitHub                         | 10    |
| 7    | 17–23 April        | <ul style="list-style-type: none"> <li>Enable Azure Monitor on VM and network resources</li> <li>Review metrics (CPU, network, disk)</li> </ul>                                                                                                                  | Document monitoring setup with screenshots              | 10    |
| 8    | 24–30 April        | <ul style="list-style-type: none"> <li>Create Azure budget alerts to track spend</li> <li>Monitor usage and adjust resource settings to stay within \$100</li> </ul>                                                                                             | Document cost management process in GitHub              | 8     |

| Week | Date     | Tasks / Deliverables                                                                                                                                                                                                                                | Submission / Evidence                        | Marks |
|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------|
| 9    | 1–7 May  | <ul style="list-style-type: none"> <li>• Create architecture diagram</li> <li>• Document full Azure environment (resources, configurations, security, monitoring)</li> </ul>                                                                        | Upload diagrams and documentation to GitHub  | 14    |
| 10   | 8–12 May | <ul style="list-style-type: none"> <li>• Prepare final presentation slides</li> <li>• Upload slides to GitHub</li> <li>• Conduct peer review (if required)</li> <li>• Submit final repository ZIP folder on Campus Online by 12 May 2026</li> </ul> | Final repository ZIP folder on Campus Online | 10    |

**Total:** [100]

**End of Assignment**